

Dallon Carlson

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Education

Wake Forest University

Ph.D. in Physics

Aug 2025 –
Winston-Salem, NC

Boise State University

B.S. in Physics, Astrophysics Emphasis

Aug 2021 – May 2025
Boise, ID

Research Experience

Graduate Research

Advisor: Dr. Caitlin Witt

Aug 2025 – Present
Wake Forest University

- **SMBHB Search Framework:** Developing an open-source, multi-band search pipeline to identify Supermassive Black Hole Binaries (SMBHBs) in variable AGN data, specifically optimized for the upcoming Vera C. Rubin Observatory (LSST).
- **Multimodal Modeling:** Expected to incorporate physical templates for relativistic Doppler boosting and variable accretion rates to distinguish periodic binary signals from stochastic intrinsic AGN variability.
- **Data Integration & Scaling:** Creating a multi-survey analysis framework that cross-correlates datasets from CRTS and LSST to extend temporal baselines and improve sensitivity to long-period systems.

Undergraduate Researcher

Advisors: Dr. Brian Jackson, Dr. Rachel Huchmala

Jan 2023 – July 2025
Boise State University

- **Exoplanet Transit Observations:** Conducted and analyzed observations of exoplanet transits using a 14" telescope and Unistellar eQuinox 2 telescopes, capturing high-precision light curves. Automated telescope operations, developed procedures for calibration (flats, darks, etc.), and resolved technical challenges like meridian flips and drift corrections.
- **Instrumentation and Calibration:** Automated telescope operations, developed procedures for calibration (flats, darks, etc.), and resolved technical challenges like meridian flips and drift corrections.
- **Data Analysis:** Processed transit data using AstrolmageJ and Python-based EXOTIC pipeline, generating light curves for exoplanetary studies. Developed technical documentation for observing procedures and trained new undergraduate researchers in observational techniques and data analysis.

Independent Research

Senior Seminar

Jan 2025 – May 2025
Boise State University

- **Formation, Dynamics, & Evolution of Hot Jupiters:** Conducted independent research into the formation, orbital dynamics, and evolution of exoplanetary systems. I specifically talked about single star, single planet systems of Hot Jupiters, and the possibilities of their formation and evolution.
- **Research Presentation:** Delivered a research talk summarizing my key findings on Hot Jupiter dynamics to a public audience.

Skills

- **Programming Languages:** Python, Java, Javascript, LaTeX
- **Astronomy Software:** AstrolmageJ, N.I.N.A, SkyX, EXOTIC (Python package for exoplanet transit reduction)
- **Data Analysis:** Handling of large datasets, statistical analysis, creating models for scientific research.
- **Observational:** Proficient in maintaining and operating small ground-based telescopes and imaging systems.
- **Research:** Writing reports, Troubleshooting errors, presenting findings in professional and academic settings.
- **Communication:** Collaboration with teams, mentoring peers, delivering conference presentations.

Publications

Doomed Worlds. I. No New Evidence for Orbital Decay in a Long-term Survey of 43 Ultrahot Jupiters Jul 2024

Elisabeth R. Adams; Brian Jackson; Amanda A. Sickafoose; Jeffrey P. Morgenthaler; Hannah Worters; Hailey Stubbers; *Dallon Carlson* et al. [10.3847/PSJ/ad3e80](https://arxiv.org/abs/10.3847/PSJ/ad3e80)

Conference Abstracts

Observing with S(u)PerP(i)g: Boise State's On-Campus Exoplanet Program Jan 2025

Dallon Carlson; Adrienne Kirk; Rachel Huchmala; Brian Jackson; Elisabeth Adams

Bulletin of the AAS, 57(2). Poster Presented at AAS 245 Winter Meeting

Observing with S(u)PerP(i)g: Boise State's On-Campus Exoplanet Program Oct 2024

Dallon Carlson; Adrienne Kirk; Rachel Huchmala; Brian Jackson; Elisabeth Adams

Bulletin of the AAS, 56(8). Poster Presented at DPS 56

Affiliations

North American Nanohertz Observatory for Gravitational Waves (NANOGrav) Feb 2026 - Present

Associate Member

Leadership and Service

Public Astronomy Observing Aug 2025 - Dec 2025
Organizer *Wake Forest University*

- Directed a series of community-focused observing sessions at Wake Forest University, under Dr. Alejandro Cárdenas-Avendaño.
- Facilitated public solar viewing using a specialized solar telescope, providing real-time explanations of solar phenomena and telescope optics to diverse audiences.

Physics Club Sept 2024 - May 2025
Vice President *Boise State University*

- Club members gained access to SPS (Society of Physics Students), increasing professional development opportunities.
- Organized regular social events to increase engagement and community within undergraduate physics students.
- Developed & implemented a telescope check-out system for club members to promote hands-on engagement in astronomy.

Teaching Experience

Astronomy Lab Instructor Aug 2025 - Present
Astronomy, Wake Forest University

Physics Lab Instructor Jan 2025 - June 2025
Physics 1 with Calculus, Boise State University